

Test Case ID	Test Area
TC-001 - View Products	Products
TC-002 - Add Product	Products
TC-003 - Search Product	Products
TC-004 - Update Product	Products
TC-005 - Update Persistence	Database / Products
TC-006 - Remove Product	Products
TC-007 - Remove Persistence	Database / Products
TC-008 - Add Customer	Customer
TC-009 - View Customer	Customers
TC-010 - Place Order	Orders
TC-011 - Insufficient Stock	Orders / Inventory
TC-012 - Invalid Quantity	Invalid Validation / Orders
TC-013 - View Sales	Sales
TC-014 - Database Persistence	Database
TC-015 - Application Exit	Application / Stability

Test Case
Verify that the user can view products stored in the system.
Verify that the user can add a new product to the system.
Verify that the user can search for an existing product by name.
Verify that the user can update the details of an existing product.
Verify that updated product information persists after restarting the application.
Verify that the user can remove an existing product from the system.
Verify that a removed product remains absent after restarting the application.
Verify that the user can add a new customer to the system.
Verify that the user can view customers stored in the system.
Verify that the user can successfully place an order for an existing customer and available product.
Verify that the system prevents an order when the requested quantity exceeds available stock.
Verify that the system rejects invalid order quantities.
Verify that the user can view recorded sales transactions.
Verify that application data is retained after closing and restarting the application.
Verify that the application exists normally without producing an error.

Preconditions
Application is running and products exist in the database.
Application is running and the product does not already exist.
Application is running and the product exists in the database.
Application is running and the product exists in the database.
A product has been successfully updated.
Application is running and the product exists in the database.
A product has been successfully removed.
Application is running and the customer does not already exist.
Application is running and at least one customer exists in the daatabase.
Application is running; customer exists; product exists; sufficient stock is available.
Application is running; customer exists; product exists; product has limited stock.
Application is running; customer and product exist.
Application is running and at least one successful order exists.
Products, customers and/or sales have been successfully stored in the system.
Application is running.

Test Steps

Launch application, Select 1. View Products

Select 2. Add Product, enter the product details, submit the product.

Select 3. Search Products, enter an existing product name - submit the search

Select 8. Update Product, enter an existing updated product details, save changes.

Update a product, close the application, restart the application, view/search the product.

Select 9. Remove Product, enter an existing product name, confirm removal.

Remove a product, close the application, restart the application, view/search the product.

Select 4. Add Customer, enter valid customer details, submit the customer.

Select 7. View Customer, review the displayed customer list.

Select 5. Place Order, enter customer, enter product, enter valid quantity, confirm order.

Select 5. Place Order, select customer, select product, enter a quantity greater than available stock, submit order.

Select 5. Place order, enter customer/product, enter an invalid quantity such as 0 or non-numeric text, submit order.

Select 6. View Sales, review the displayed sales records.

Add/update/store data, close the application normally, restart the application, check the stored data.

Launch the application, Select 10. Exit, observe the application response.

Test Data
Existing products
A new product with valid name, size, price, color, style and stock.
An existing product, e.g. Black Hoodie.
Existing product with modified details.
The product updated during TC-004.
An existing product intended for removal.
The product removed during TC-006.
New customer's name and email address.
Existing customers in the database.
Customer: Bhlomingtn; Product: Black Hoodie; Quantity: 2
Grey Beanie - request 10 when only 8 are available.
Invalid input and the resulting validation message.
Previously completed Black Hoodie order.
Previously created or updated product/ customer/sale records.
Exit option 10.

Expected Result
Products are displayed with their relevant details.
Product is successfully added and appears in the product inventory.
The matching product is found and its details are displayed correctly.
Product details are successfully updated and the new information is displayed.
The updated product details remain unchanged after restarting the application
The product is successfully removed and no longer appears in the product inventory.
The removed product does not reappear after the application is restarted.
Customer is successfully added and stored in the database.
The system displays the stored customers and their relevant details correctly.
Order successfully processed, receipt/order details are generated, and the transaction is recorded.
The order is rejected and system informs the user that there is insufficient stock. Inventory is not reduced incorrectly.
Screenshot showing the invalid input and the resulting validation message.
The system displays recorded sales with the relevant customer, product, quantity, total and transaction details.
Previously saved data remains available after restarting the application.
The application closes normally without displaying an error or traceback.

Actual Result	Status
Products displayed succesfully.	PASS
New product is added successfully.	PASS
Product details displayed successfully.	PASS
Successful update message.	PASS
	PASS
The updated product shows after restarting the application.	PASS
The removed product remains absent after restart the application.	PASS
The customer being successfully added.	PASS
The customer list displayed by the system.	PASS
Succesful order/receipt showing customer, product, quantity, price and total.	PASS
Insufficient stock message and rejected order.	PASS
Quantity must be greater than zero. / Invalid quantity. Please enter a number.	PASS
Record sales displayed correctly.	PASS
Previously saved data still available after application restart.	PASS
Normal application exit.	PASS

Evidence ID
Screenshot 2026-09-18 155359.png
Screenshot 2026-09-18 155516.png / Screenshot 2026-09-18 155625.png
Screenshot 2026-09-18 155717.png
Screenshot 2026-09-18 155625.png / Screenshot 2026-09-18 160549.png
Screenshot 2026-09-18 155717.png
Screenshot 2026-09-18 160736.png
Screenshot 2026-09-18 160914.png
Screenshot 2026-09-18 155832.png
Screenshot 2026-09-18 160225.png
Screenshot 2026-09-18 160031.png
Screenshot 2026-09-18 181940.png
Screenshot 2026-09-18 181447 / Screenshot 2026-09-18 181611.png / Screenshot 2026-09-18 181611.png
Screenshot 2026-09-18 160122.png
Screenshot 2026-09-18 182224.png
Screenshot 2026-09-18 160811.png

Evidence Description
Screenshot showing the product list displayed successfully.
Screenshot showing the new project successfully added and displayed in inventory.
Screenshot showing the search being performed and matching product details displayed.
Screenshot showing the product before/after the update and the successful update message.
Screenshot showing the updated product after restarting the application.
Screenshot showing the product removal and confirmation that it was removed.
Screenshot showing that the removed product remains absent after restarting the application.
Screenshot showing the customer being successfully added.
Screenshot showing the customer list displayed by the system.
Screenshot of the successful order/receipt showing the customer, product, quantity, price and total.
Screenshot showing the insufficient-stock message and rejected order.
Screenshots showing invalid quantity or invalid validation message.
Screenshot showing the recorded sale displayed correctly.
Screenshot showing previously saved data still available after application restart.
Screenshot showing the application exit message/normal termination, if available.